

# Text carries more than words

A single message is ambiguous by construction. The same string can mean opposite things, and only the conversation tells you which.

**“Fine.”**

relieved · neutral · cold · passive-aggressive

**73%**

of probe utterances flip  
sentiment with history

**10,956**

messages analysed in  
the published corpus

**4**

platforms parsed natively

# Message-level analysis misses the conversation

We measured it rather than asserting it. On a controlled probe set where the wording is identical and only the history differs, every context-free reader collapses to the same score — the majority-class rate.

**0.636**

text-only ceiling (analytic)

**0.636**

TF-IDF text-only

**0.636**

frozen MiniLM  
transformer, no context

The design balances every utterance across its gold labels, so the mutual information between the words and the label is zero. A pretrained encoder buys nothing here: it is bounded by the same ceiling.

### 3 · OUR SOLUTION

# CEREBRO

A context-aware, temporal conversation-intelligence engine. It does not ask *what is the sentiment of this message*. It asks *what is happening to the emotional state of this conversation, where did it change, and what accompanied the change*.

**0.909**

probe accuracy with  
context

**+27.3 pp**

over the text-only ceiling

**p = 0e+00**

exact McNemar vs text-  
only

95% bootstrap CI on the paired gain: [0.210, 0.347] accuracy.

# Architecture

INPUT · PS-01 \$2



### CHAT PARSER + AUTO-DETECT

one common format · speakers · timestamps

### PREPROCESSING + BEHAVIORAL FEATURES

16-dim vector · CAPS · exclamations · emoji · response gap

### CONTEXT + SPEAKER MEMORY

sliding 4-turn window · decayed summary · per-speaker state

### MESSAGE REPRESENTATION

text ⊕ context ⊕ behavior ⊕ memory

### MULTI-TASK NLP ENGINE

sentiment · emotion · tone · tension (0-100)

### HIDDEN-SIGNAL DETECTION

sarcasm · irony · passive-aggression · learned ⊕ contradiction

### TEMPORAL ENGINES

UNDERSTANDING · \$5 \$8 \$9 \$17

TELLIGENCE · \$10-22

arc · transitions · turning points · escalation trajectory



### MODEL FUSION + CALIBRATION

six evidence streams · validation-driven weights · Platt scaling



### EXPLAINABILITY + CONVERSATION REPORT

WHY? · WHAT CHANGED? · speaker profiles · 18-section report



### EMOTIONAL ARC · TENSION CURVE

the journey, explained — served by the FastAPI dashboard

End-to-end pipeline: parse → features → contextual multi-task heads → hidden signals → temporal engines → fusion → explainability → API + dashboard.

# Context, memory, behaviour — and a proof they are used



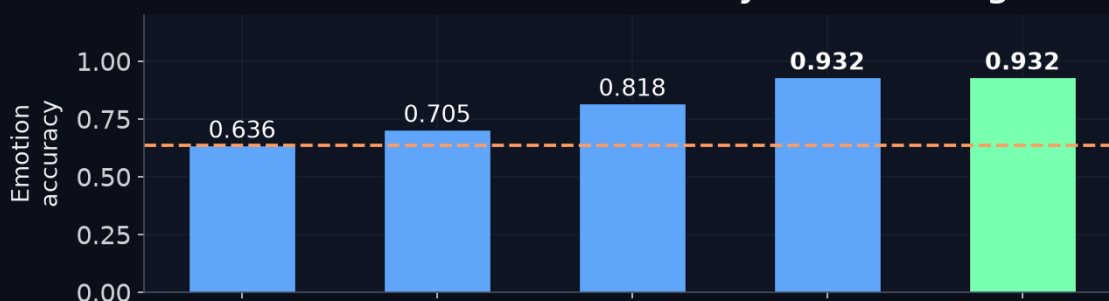
## Controlled proof — context is required, not merely helpful

176 probe turns · identical wording under benign vs tense histories · balanced so  $I(\text{text}; \text{label})=0$

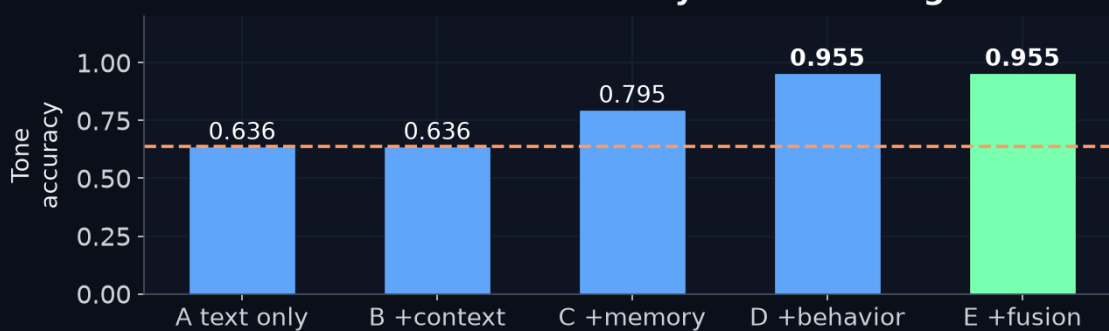
**Sentiment — dashed line = text-only lexical ceiling 0.636**



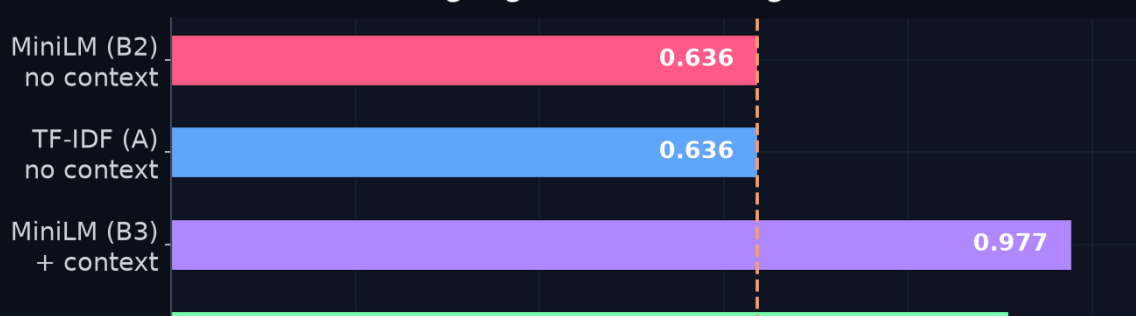
**Emotion — dashed line = text-only lexical ceiling 0.636**



**Tone — dashed line = text-only lexical ceiling 0.636**



**No amount of language understanding substitutes for context**





CEREBRO full (E)



0.0

0.2

0.4

0.6

0.8

1.0

sentiment accuracy on probe turns (held-out history wording)

0.909

The component ladder on held-out history wording. Text-only sits exactly on the lexical ceiling; the full stack reaches 0.909 sentiment / 0.932 emotion / 0.955 tone.

PS-01 §11, §12, §36

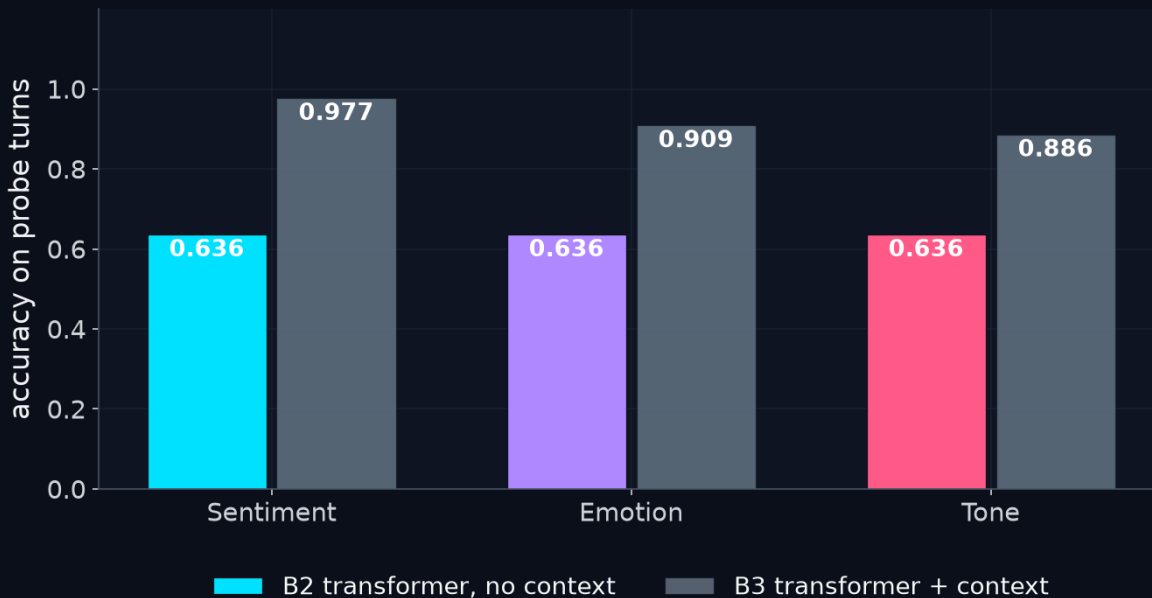
# Baselines: lexical, and a real pretrained transformer



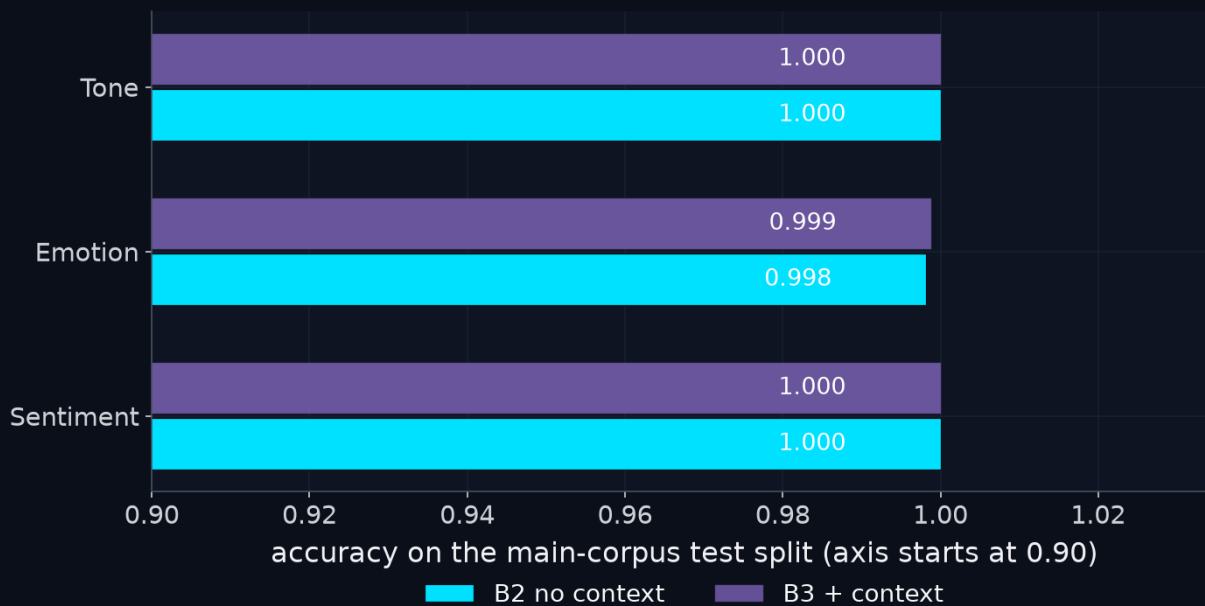
## Pretrained-transformer baselines B2 and B3

frozen sentence-transformers/all-MiniLM-L6-v2 (ONNX, CPU) + the same heads — no fine-tuning, so the comparison isolates context

### A pretrained encoder still needs the conversation around it



### The template corpus saturates every model family alike



B2 is a frozen sentence-transformers MiniLM encoder — a strong, general-purpose language model with no fine-tuning, exactly as §9 asks. Bottom: the template corpus saturates every model family alike, which is why the probe experiment is the informative one.

PS-01 §9, §36

# Every number measured, every claim bounded

| HEAD               | METRIC                                 | VALUE                 |
|--------------------|----------------------------------------|-----------------------|
| Sentiment          | probe accuracy (unseen wording)        | 0.909                 |
| Emotion            | probe accuracy (unseen wording)        | 0.932                 |
| Tone               | probe accuracy (unseen wording)        | 0.955                 |
| Sarcasm            | ROC-AUC (held-out test)                | 0.9485                |
| Irony              | ROC-AUC                                | 0.9267                |
| Passive aggression | ROC-AUC                                | 0.9386                |
| Tension            | MAE / R <sup>2</sup>                   | 3.205 / 0.975         |
| Calibration        | top-1 ECE (sentiment / emotion / tone) | 0.006 / 0.017 / 0.019 |

In-corpus scores saturate because the corpus is template-generated (62 distinct phrasings over 10,956 messages). We publish that rather than headlining it, and report real-text transfer separately.

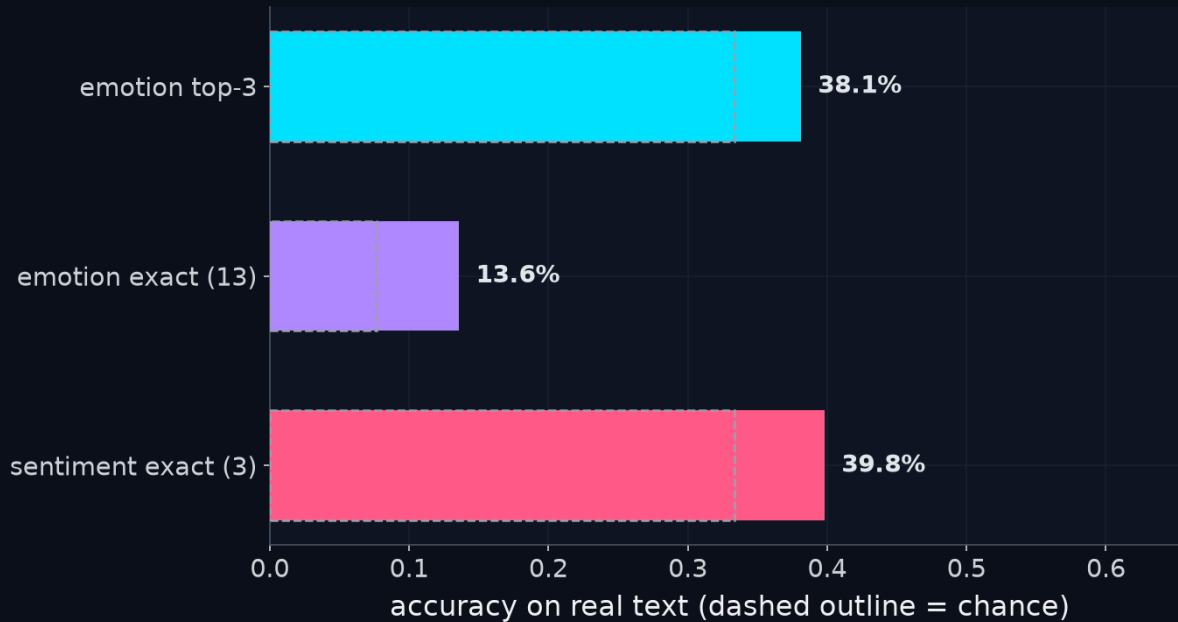
# Real human text is the honest test



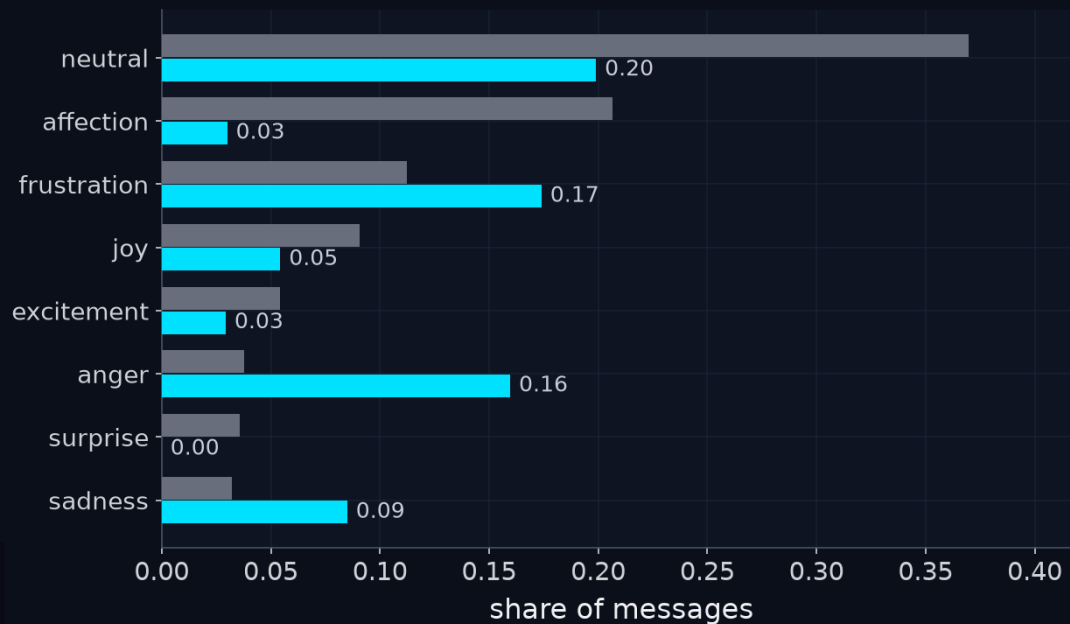
## Zero-shot transfer to real GoEmotions text

3,000 real Reddit comments · honest cross-corpus metrics (not comparable to in-corpus tables)

### Zero-shot accuracy vs chance



### Label shift — grey = gold, cyan = predicted



Zero-shot on GoEmotions, then the same heads fine-tuned on real Reddit text — emotion exact

0.136 → 0.446, top-3 → 0.753.

PS-01 §35 — measured on google-research-datasets/go\_emotions (simplified), research license (research licence)

# Explainable, robust, fast

**176**

probe turns in the  
controlled experiment

**88**

messages / second,  
single CPU core

**8.9 MB**

peak memory, 1,000-  
message chat

**184 ms**

API round-trip, 50  
messages

**0.92**

min annotator  $\kappa$  across  
heads

**20/20**

robustness scenarios  
passed

Data-leakage audit: 0 conversation overlap across splits, 0 label-leakage findings; the 62 recurring templates are disclosed, not hidden. Every prediction ships with confidence and a human-readable explanation that separates detected evidence from inferred interpretation.



# Where this goes

## Applications

- Customer-support escalation early-warning
- Team-communication and workplace analytics
- Community moderation triage
- Conversation research at scale

## Next

- Multilingual parsing and lexicons
- Multi-turn real conversation corpora
- Streaming / real-time analysis
- Human multi-annotator agreement study

**“Decode the shift. Understand the conversation.”**

CEREBRO reports *model-detected emotional and tone signals*, not a person's true feelings. Turning points are statistical associations, not causal claims.